TIANCHONG JIANG

tianchongj@ttic.edu Homepage GitHub
Google Scholar Chicago, IL

Education

Toyota Technological Institute at Chicago Sep. 2024 – Jun. 2028 (Expected)
PhD in Computer Science
Advisor: Matthew R. Walter

University of Chicago Sep. 2019 – Dec. 2023
B.A. Mathematics | B.S. Computer Science with Honors
GPA: 3.89/4.00; magna cum laude

Publications and Preprints

T. Jiang*, R. Xu*, Y. Li, X. Tan, and M. R. Walter. "Real-to-Sim-to-Real at the Last Millimeter," in *Conference on Robot Learning (CoRL) Late-Breaking Results*, 2026.

Ludo Robotics (incl. **T. Jiang**). "Ludi$_{0.1}$: An Agentic System for Socially Intelligent Robots," *Technical report, arXiv:2608.22035*, 2026.

T. Jiang, X. Tan*, S. W. Wheeler*, L. Sun*, T. W. Ayalew*, and M. R. Walter. "What Are We Actually Benchmarking in Robot Manipulation?," in *Conference on Robot Learning (CoRL)*, 2026. **Oral**.
Also at the *IROS 2026 RGMCW Workshop*: **RGMCW Workshop Best Abstract Award**.

T. Jiang, J. Ji, X. Tan, J. Fang, A. Bhattad, V. Guizilini†, and M. R. Walter†. "Do You Know Where Your Camera Is? View-Invariant Policy Learning with Camera Conditioning," in *2026 IEEE International Conference on Robotics and Automation (ICRA)*, 2026. **Best Paper Award on Robot Learning**.

X. Tan, J. Ji, **T. Jiang**, P. Lopes, and M. R. Walter. "HapCompass: A Rotational Haptic Device for Contact-Rich Robotic Teleoperation," in *2026 IEEE International Conference on Robotics and Automation (ICRA)*, 2026.

L. Sun*, T. Yoneda*, S. W. Wheeler, **T. Jiang**, and M. R. Walter. "StackGen: Generating Stable Structures from Silhouettes via Diffusion," in *2025 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2025.

T. Ayalew, X. Zhang*, K. Y. Wu*, **T. Jiang**, M. Maire, and M. R. Walter. "PROGRESSOR: A Perceptually Guided Reward Estimator with Self-Supervised Online Refinement," in *2025 IEEE/CVF International Conference on Computer Vision (ICCV)*, 2025.

T. Yoneda*, J. Fang*, P. Li*, H. Zhang*, **T. Jiang**, B. Picker, D. Yunis, S. Lin, H. Mei, and M. R. Walter. "Statler: STATe-maintaining Language models for Embodied Reasoning," in *2024 IEEE International Conference on Robotics and Automation (ICRA)*, 2024.

T. Yoneda*, **T. Jiang***, G. Shakhnarovich, and M. R. Walter. "6-DoF Stability Field via Diffusion Models," *arXiv preprint*, 2023.

Experience

AI/ML Research Intern, Ludo Robotics / KRAFTON June 2026 – August 2026
• Worked on sim-to-real transfer for a humanoid robot.

Research Assistant, Robot Intelligence through Perception Lab October 2024 – Present
• Studying view-invariant imitation learning with camera conditioning and Plucker ray representations for robust visuomotor policies.
• Auditing robot manipulation benchmarks for shortcut solvability, statistical fragility, creeping overfitting, and data-source dependence.

Research Assistant, Robot Intelligence through Perception Lab August 2022 – June 2024
• Implemented an SE(3) diffusion model to predict stable object configurations. (MuJoCo, PyTorch)
• Co-developed a Large Language Model interface for a UR5 robot arm, demonstrated to thousands of visitors at the Museum of Science and Industry in Chicago. (ROS, UR5)

Teaching

TA for MPCS 50103 Discrete Mathematics — Autumn 2022
TA for TTIC 31170 Planning, Learning, and Estimation for Robotics and Artificial Intelligence — Spring 2025

Service

Reviewer for IROS 2025; ICCV 2025; ICRA 2026; CoRL 2026; ACL ARR 2026; AAAI 2027; TMLR

Skills

ROS, MuJoCo, Isaac Sim, PyTorch, Docker, LaTeX, Git, Slurm
Robot Platforms: UR5, Unitree G1, Hiwonder NexArm, Koch Arms, SO-101 Arms, ARCTOS, TurtleBot

Relevant Coursework

Introduction to Machine Learning; Algorithms; Algorithmic Game Theory; Fundamentals of Deep Learning; Generative Models, Art and Perception; Planning, Learning, and Estimation for Robotics and Artificial Intelligence; Unsupervised Learning; Mathematical Toolkit; Learning on Graphs and Manifolds; Introduction to Computer Vision; Introduction to Statistical Machine Learning